TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Water Repellent and Flame-retardant Effects

Two pad–dry–cure productions on cotton — a low-energy fluorocarbon film that sheds water, and a phosphorus finish that stops cotton from sustaining flame.

WP-4202 · TEXTILE FINISHING SESSIONAL · EXPERIMENT 1 · SPECIALIZATION — WET PROCESSING

1. Aim and theory

Aim: to produce (A) a durable **water-repellent** finish on cotton (and, for comparison, on polyester) with a fluorocarbon emulsion, and (B) a durable **flame-retardant (FR)** finish on cotton with a Pyrovatex CP-type organophosphorus product, both by the pad–dry–cure route; to calculate the chemical add-on in each case from bath concentration and wet pick-up; and to carry out a first quick assessment — spray/beading behaviour for repellency and char behaviour for flame retardancy. Full standardised testing of these same fabrics is the subject of Experiment 4.

Water repellency is a surface-energy effect, not waterproofing. Water (surface tension about 72 mN/m) beads and rolls off a surface whose critical surface energy is lower than that of water, because it cannot spread and wet the fibres. The finish therefore coats each fibre with a thin film presenting $–CF_2–$ and terminal $–CF_3$ groups outward. Fluorocarbons give the lowest surface energies of any textile finish (about 10–15 mN/m for a well-oriented $–CF_3$ surface), which is why they repel oil as well as water. Silicone and hydrocarbon (wax/resin) repellents are cheaper alternatives, but silicones repel water only (not oil) and waxes are far less durable to washing. In all cases the fabric pores stay open: the garment still breathes, and water under pressure will still pass — that is the distinction between *water-repellent* and *waterproof* (coated or laminated).

The modern context must be stated honestly. Long-chain (C8) fluorocarbons have been withdrawn in most markets because their breakdown products (PFOA-type) persist in the environment. Current commercial products are short-chain C6 fluorocarbons or, increasingly, fluorine-free (C0) dendrimer / silicone-hybrid repellents, which give good water repellency but little or no oil repellency. The pad–dry–cure mechanics taught in this experiment are identical for all three families; only the emulsion in the trough changes.

Flame retardancy is a combustion-chemistry effect. Untreated cotton ignites readily, burns with a flame, and continues to glow — its limiting oxygen index (LOI) is only about 18.4%, below the 21%

oxygen of air. Cellulose on heating depolymerises to volatile, flammable levoglucosan, which feeds the flame. A durable phosphorus finish for cotton works mainly in the *condensed phase*: on heating, the phosphorus compound releases phosphoric / polyphosphoric acid, which phosphorylates cellulose and catalyses its dehydration to **char** (carbon + water) instead of levoglucosan. Char is a poor fuel and an insulating barrier, so the flame starves. Pyrovatex CP-type products are N-methylol phosphonopropionamides: the N-methylol (–NH–CH_2OH) group lets the molecule react with cellulose –OH groups and self-crosslink during curing, in the presence of a resin co-reactant and an acid catalyst, which is what makes the finish survive repeated laundering. Nitrogen in the molecule acts synergistically with phosphorus (the P–N synergism), raising efficiency.

KEY IDEA — TWO OPPOSITE SURFACE JOBS

Water repellency changes only the fibre *surface* (a film a few nanometres thick is enough). Flame retardancy must change the *bulk decomposition* of the fibre, which is why FR add-ons (here 17.5% owf) are roughly nine times larger than repellent add-ons (here 1.95% owf). Never judge a finish by bath concentration alone — judge it by add-on on the fabric.

2. Apparatus, materials and safety

ITEM	SPECIFICATION / USE
Laboratory padder	Two-bowl horizontal padder with adjustable nip pressure; trough capacity about 300 mL
Stenter / curing oven	Pin frame or laboratory stenter capable of 110 °C drying and 150–170 °C curing, with timer
Fabric	Desized, scoured and bleached cotton, about 150 g/m², pieces 30 cm × 20 cm (about 9 g each); one polyester piece of similar size for the repellent comparison
Chemicals A	Fluorocarbon repellent emulsion (C6 type); wetting agent (non-rewetting type); acetic acid for pH adjustment
Chemicals B	Pyrovatex CP-type FR product; trimethylolmelamine (TMM) or etherified melamine resin co-reactant; phosphoric acid (85%) catalyst; urea (optional buffer); non-ionic wetting agent
Assessment kit	Spray-test apparatus or wash bottle and distilled water, white blotting paper; Bunsen burner and tongs for a supervised char-behaviour check only

General	Balance (±0.01 g), beakers, measuring cylinders, pH paper or meter, thermometer, stopwatch, scissors, marker

Safety. Wear gloves and eye protection for the FR bath: it is acidic and the product contains reactive N-methylol groups that can release small amounts of formaldehyde during curing — cure in a ventilated oven and do not lean over the fabric as it leaves the stenter. The char-behaviour check in Section 6 is a lecturer-supervised demonstration over a sink, with tongs, never a hand-held test. Fluorocarbon and FR baths are not poured to the drain neat; collect residues for the laboratory waste procedure.

3. The pad–dry–cure line

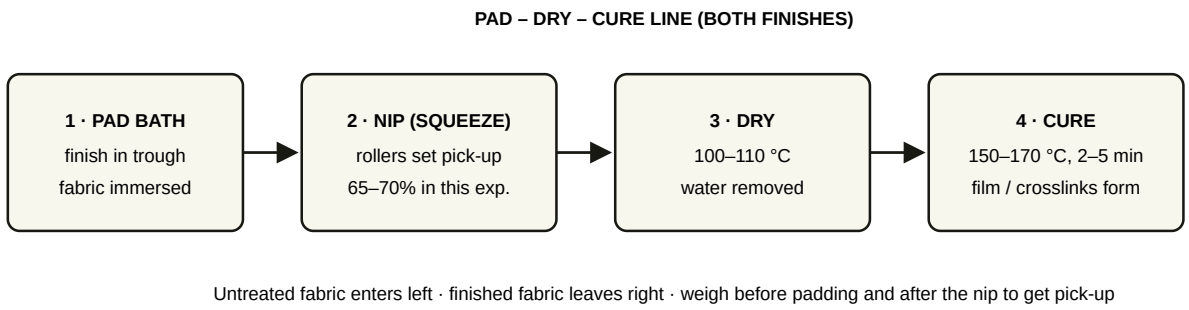


Fig. 1 — Pad–dry–cure line used for both productions. The nip, not the trough, decides how much liquor — and therefore how much chemical — the fabric carries.

Wet pick-up (WPU) is defined as:

$$\text{Wet pick-up (\%)} = [(\text{wet mass after nip} - \text{dry mass}) \div \text{dry mass}] \times 100$$

and the dry chemical add-on follows directly:

$$\text{Add-on (g/kg fabric)} = \text{bath concentration (g/L)} \times \text{wet pick-up (as a fraction)}$$

because 1 L of bath carried per kilogram of fabric at 100% pick-up delivers the bath concentration in grams per kilogram. This single relation is used for every calculation in this course.

4. Production A — fluorocarbon water-repellent finish

Recipe A

COMPONENT	AMOUNT	FUNCTION
Fluorocarbon repellent emulsion (C6)	30 g/L	Low-surface-energy film former

Wetting agent (non-rewetting)	1 g/L	Rapid, even wetting in the trough without harming repellency later
Acetic acid (to pH)	pH 5.0–5.5	Bath stability; many repellent emulsions are cationic and pH-sensitive
Wet pick-up target	65%	Set by nip pressure; verify by weighing

Procedure A

1. Condition and weigh each dry piece (cotton and polyester). Record the dry mass.
2. Make up 300 mL of Recipe A: add the wetting agent to water first, adjust pH, then stir in the fluorocarbon emulsion gently — vigorous stirring foams the bath and can break the emulsion.
3. Pass each piece through the bath (immersion about 15 s) and through the nip **twice** (pad–pad) for even distribution. Weigh immediately: adjust nip pressure until the wet pick-up is 65% ± 3%.
4. Dry at 100–110 °C for 2–3 min until just dry. Do not over-dry before curing.
5. Cure at 150 °C for 3 min (cotton) / 150 °C for 2 min (polyester). Curing melts and orients the fluorocarbon side chains so the $–CF_3$ groups face outward — an uncured sample repels poorly even at the same add-on.
6. Condition for at least 4 h (overnight is better) before assessment: repellency develops fully as the film relaxes. Label each piece with recipe, pick-up and cure.

Worked calculation A — add-on

$$\text{Add-on} = 30\ \text{g/L} \times 0.65 = 19.5\ \text{g/kg fabric} \ (= 1.95\%\ \text{owf})$$

For a 9.0 g cotton piece this is 19.5 × 0.009 = 0.176 g of repellent solids on the piece — a reminder of how thin a functional surface film needs to be. If the measured pick-up comes out at, say, 58% instead of 65%, the true add-on is 30 × 0.58 = 17.4 g/kg, and the record sheet must show the measured value, not the target.

Note on polyester. The same bath and conditions are used for the polyester comparison piece. Polyester is already hydrophobic (moisture regain about 0.4%), so its untreated beading looks deceptively good; the finish still improves spray rating and, importantly, gives oil repellency and soil resistance that the bare fibre lacks. On polyester the film adheres physically rather than through fibre –OH groups, so wash durability is typically lower than on cotton unless a blocked-isocyanate extender/crosslinker is added.

5. Production B — durable flame-retardant finish (Pyrovatex CP type)

Recipe B

COMPONENT	AMOUNT	FUNCTION
Pyrovatex CP-type product	250 g/L	Phosphorus FR agent; reacts with cellulose via its N-methylol group
Melamine resin (TMM type)	30 g/L	Co-reactant / crosslinker; supplies extra nitrogen (P–N synergism) and durability
Phosphoric acid (85%)	15 g/L	Acid catalyst for the cure reaction
Urea	10 g/L	Buffer / formaldehyde scavenger; softens handle slightly
Non-ionic wetting agent	2 g/L	Even penetration of the heavy bath
Wet pick-up target	70%	High pick-up is deliberate — FR needs a high add-on

Procedure B

1. Weigh the dry cotton pieces. Make up 300 mL of Recipe B in the order listed, stirring the FR product in last-but-one and the acid last. Check the pH: the bath should be distinctly acidic (about pH 3–4).
2. Pad twice at a nip setting giving 70% wet pick-up; verify by weighing.
3. Dry at 110 °C for 3 min. The dried fabric is still essentially untreated chemically — the reaction has not yet happened.
4. Cure at 160–170 °C for 3–4 min. This is the step in which the N-methylol groups condense with cellulose –OH and with the melamine resin, locking the phosphorus onto the fibre. Under-curing gives a finish that washes out; over-curing yellows and weakens cotton.
5. After-wash: rinse warm with 2 g/L soda ash solution, then in water. This neutralises residual acid, removes unreacted product and reduces free formaldehyde. Dry and condition.
6. Feel the handle and note it: a durable FR finish at this add-on stiffens cotton noticeably and typically costs some tensile and tear strength (commonly of the order of 15–25% if curing is harsh). Record any yellowing.

Worked calculation B — add-on

$$\text{Add-on} = 250\ \text{g/L} \times 0.70 = 175\ \text{g/kg} = 17.5\%\ \text{owf}$$

On a 9.0 g piece that is 1.58 g of FR product — compare 0.176 g in Production A. The phosphorus content actually fixed is a fraction of this (the commercial product is not pure phosphorus), which is why the recipe also carries resin and why cure control matters more than simply raising the bath concentration. State the add-on on every record as product add-on (17.5% owf), never as “17.5% phosphorus”.

6. Quick assessment (first look only)

CHECK	METHOD	EXPECTED RESULT
Beading (A)	Drops of distilled water placed from 1 cm height	Distinct spherical beads standing on the surface, not spreading within 30 s; untreated cotton wets instantly
Spray shower (A)	Water poured/sprayed over fabric held at 45°	Water runs off in streams; fabric underneath largely dry; blotting paper behind stays near-dry. Compare cotton vs polyester, treated vs untreated
Char behaviour (B)	Supervised edge-ignition demonstration: 2 s flame contact, then remove	Untreated cotton: continues to flame and glow to ash. FR cotton: flame dies on removal (self-extinguishing), leaving a coherent black char that holds the fabric shape
Handle / appearance	Hand feel + visual	A: almost unchanged. B: stiffer, sometimes slightly yellow — both are real costs of the finish and must be recorded, not hidden

These are screening observations. They tell you the finish is present; they do not give a grade that can be quoted to a buyer. Experiment 4 repeats the assessment with the standard spray rating, the vertical flame test and LOI concepts, on fabrics from this experiment.

7. Results (as recorded — template with typical values)

PRODUCTION	PICK-UP (%)	ADD-ON	QUICK ASSESSMENT
A — cotton + fluorocarbon	65	19.5 g/kg (1.95% owf)	Strong beading; spray run-off clean

A — polyester + fluorocarbon	65 (set)	19.5 g/kg nominal	Strong beading; untreated control also beads but wets through under spray
B — cotton + Pyrovatex type	70	175 g/kg (17.5% owf)	Self-extinguishing in demo; intact char; handle stiffened
Control — untreated cotton	—	—	Wets instantly; burns completely (LOI ≈ 18.4%)

LOI REFERENCE — KEEP FOR EXPERIMENT 4

Untreated cotton LOI ≈ **18.4%**. Effectively FR-treated cotton typically reaches LOI ≈ **27–30%** (typical range). Because air is 21% oxygen, an LOI above about 26% is a useful rule-of-thumb marker of a fabric that will not sustain burning in normal air — but LOI is a laboratory index, not a guarantee of safety in a real fire.

8. Precautions

- Verify pick-up by weighing for *every* piece. A recipe quoted without its pick-up cannot be reproduced.
- Add acid to the FR bath last and never store the mixed catalysed bath overnight — it slowly self-condenses and loses reactivity.
- Respect cure windows. Five extra minutes at 170 °C does not give “extra durability”; it gives tendered, yellowed cotton.
- After-wash the FR fabric before handling it extensively or testing it, otherwise unreacted product inflates both the add-on and the first test result.
- Condition finished pieces before assessment; testing a warm, freshly cured repellent fabric understates its performance.

9. Practice (viva preparation)

1. Why does a water-repellent fabric still let water through under pressure, and what product would you choose instead if waterproofing were required?
2. Calculate the add-on if the fluorocarbon bath is raised to 40 g/L but the nip is opened so pick-up falls to 55%. Has the finish on the fabric increased or decreased?
3. Explain the condensed-phase action of a phosphorus FR finish on cotton, naming the product of cellulose decomposition that is suppressed and the product that is promoted.
4. Why is the FR add-on (17.5% owf) so much larger than the repellent add-on (1.95% owf)?

5. What happens to durability in each production if the cure step is omitted, and why do the two cases differ?
6. State the LOI of untreated cotton and the typical LOI range after effective FR treatment, and say what the numbers mean in air.

Answer key

1. Repellency is only a fibre-surface energy effect; the pores remain open, so hydrostatic pressure forces water through. For waterproofing use a continuous film: a coated (e.g. PU-coated) or laminated fabric.
2. Add-on = 40 × 0.55 = 22.0 g/kg. It has increased (from 19.5 g/kg), but only modestly despite a 33% rise in bath concentration, because pick-up fell. Bath concentration alone never states the finish level.
3. Phosphoric acid released on heating phosphorylates cellulose and catalyses dehydration to char + water, suppressing formation of volatile levoglucosan that would otherwise feed the flame. The char insulates the underlying fibre.
4. Repellency needs only a nanometre-scale surface film; flame retardancy must redirect the decomposition of the whole fibre mass, which requires phosphorus distributed throughout the fabric at high loading.
5. Repellent: the emulsion dries as a loose film with poorly oriented side chains — repellency is weak and washes off quickly. FR: without cure the N-methylol condensation with cellulose never occurs, so essentially all product washes out in the after-wash; durability is lost almost completely. In both cases cure is what converts a deposit into a finish, but the FR case depends on a covalent reaction, so the loss is total.
6. Untreated cotton ≈ 18.4%; FR-treated cotton typically 27–30%. Air contains 21% oxygen, so untreated cotton burns readily in air, whereas a fabric needing 27–30% oxygen to sustain burning tends to self-extinguish in air.
